

Supporting Information for Pwned: The Risk of Exposure From Data Breaches*

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*Data and scripts behind the analysis presented here can be downloaded from <http://github.com/themains/pwned>.

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SI 1 Tables

The missing entries reflect cases where we do not have commensurate categories in our data. The largest differences we see are on race and ethnicity, a variable where our coding differs from CPS in meaningful ways.

All analyses retain the 5000 sampled respondents. Counts are distinct person–breach pairs for the one email queried per respondent. The primary outcome includes all recorded exposures; the secondary outcome includes only verified, nonspam records and assigns zero to respondents with none. Ages 18–25 include the lower endpoint. Race labels follow the supplied YouGov codebook.

The regression tables report unadjusted OLS estimates, standard errors, and 95% confidence intervals, retaining the historical specification. Reference groups are White, Female, and No HS; the age model uses a natural spline with two degrees of freedom. HC3 sensitivity estimates and Welch sex contrasts for both outcomes are provided in the generated CSV results. Intervals describe respondent-level sampling variation under independence; they do not account for panel selection or incomplete HIBP coverage. Group plots retain the original mean ± 1.96 standard-error intervals; the age plot retains the loess fit and its confidence band.

Table SI 1.1: Comparison Between YouGov and CPS 2018

	CPS	YouGov	Difference
Age			
18 to 25	0.14	0.15	-0.01
26 to 35	0.18	0.18	0.00
36 to 50	0.25	0.23	0.01
51 to 65	0.25	0.26	-0.01
66 to 80+	0.19	0.18	0.01
Sex			
Male	0.48	0.49	0.00
Female	0.52	0.51	0.00
Race			
White alone	0.78	0.64	0.13
Black or African American alone	0.13	0.12	0.01
American Indian and Alaska Native alone	0.01	0.01	0.00
Asian alone	0.06		
Native Hawaiian and Other Pacific Islander alone	0.00		
Two or more races	0.02		
Educational Attainment			
No high school diploma	0.11	0.07	0.04
High school or equivalent	0.29	0.33	-0.05
Some college, less than 4-yr degree	0.28	0.31	-0.03
Bachelor's degree or higher	0.32	0.29	0.03

Table SI 1.2: Recorded Exposures by Race/Ethnicity

Term	Estimate	SE	Lower	Upper	p
(Intercept)	3.113	0.046	3.024	3.203	0.000
Black	0.044	0.116	-0.184	0.271	0.707
Hispanic/Latino	-0.618	0.104	-0.822	-0.415	0.000
Asian	-0.290	0.227	-0.735	0.155	0.202
Native American	-0.152	0.363	-0.863	0.559	0.676
Mixed Race	-0.457	0.234	-0.915	0.001	0.051
Other	-0.667	0.288	-1.233	-0.102	0.021
Middle Eastern	-0.190	0.721	-1.603	1.223	0.792

Table SI 1.3: Recorded Exposures by Sex

Term	Estimate	SE	Lower	Upper	p
(Intercept)	3.157	0.051	3.057	3.258	0
Male	-0.344	0.074	-0.488	-0.199	0

Table SI 1.4: Recorded Exposures by Education

Term	Estimate	SE	Lower	Upper	p
(Intercept)	2.349	0.142	2.071	2.627	0.000
HS Grad.	0.534	0.155	0.229	0.839	0.001
Some College	0.683	0.164	0.361	1.004	0.000
2-year College Degree	0.719	0.180	0.367	1.072	0.000
4-year College Degree	0.853	0.166	0.528	1.178	0.000
Postgrad Degree	0.851	0.180	0.498	1.205	0.000

Table SI 1.5: Recorded Exposures by Age

Term	Estimate	SE	Lower	Upper	p
(Intercept)	1.989	0.091	1.810	2.167	0
Age spline 1	2.281	0.203	1.884	2.679	0
Age spline 2	-0.900	0.209	-1.309	-0.491	0

SI 2 Figures

Figure SI 2.1: Relationship Between Race and Number of Breaches

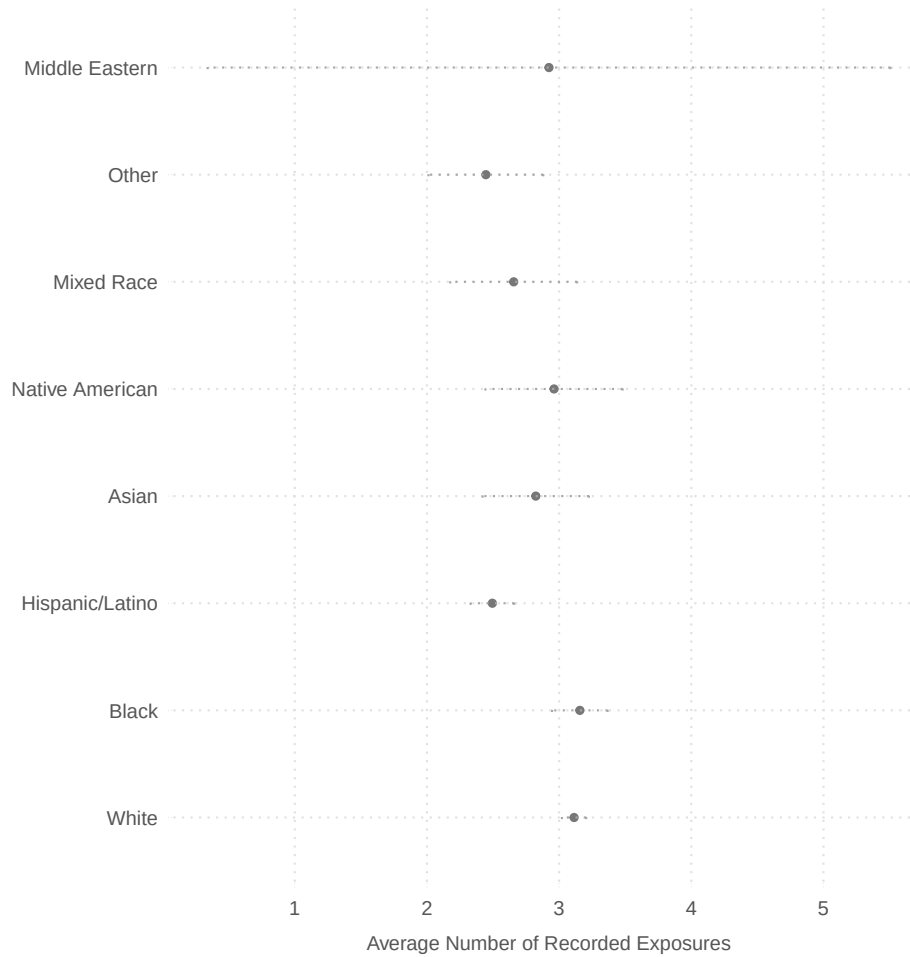


Figure SI 2.2: Relationship Between Sex and Number of Breaches

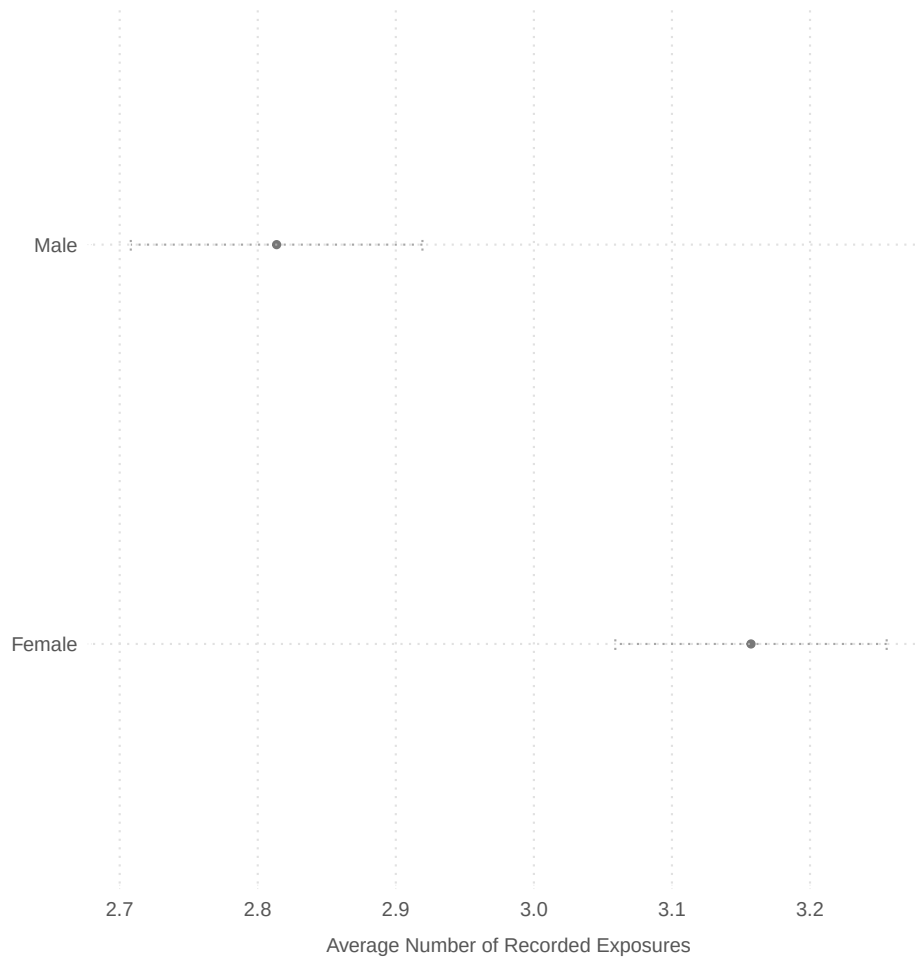


Figure SI 2.3: Relationship Between Education and Number of Breaches

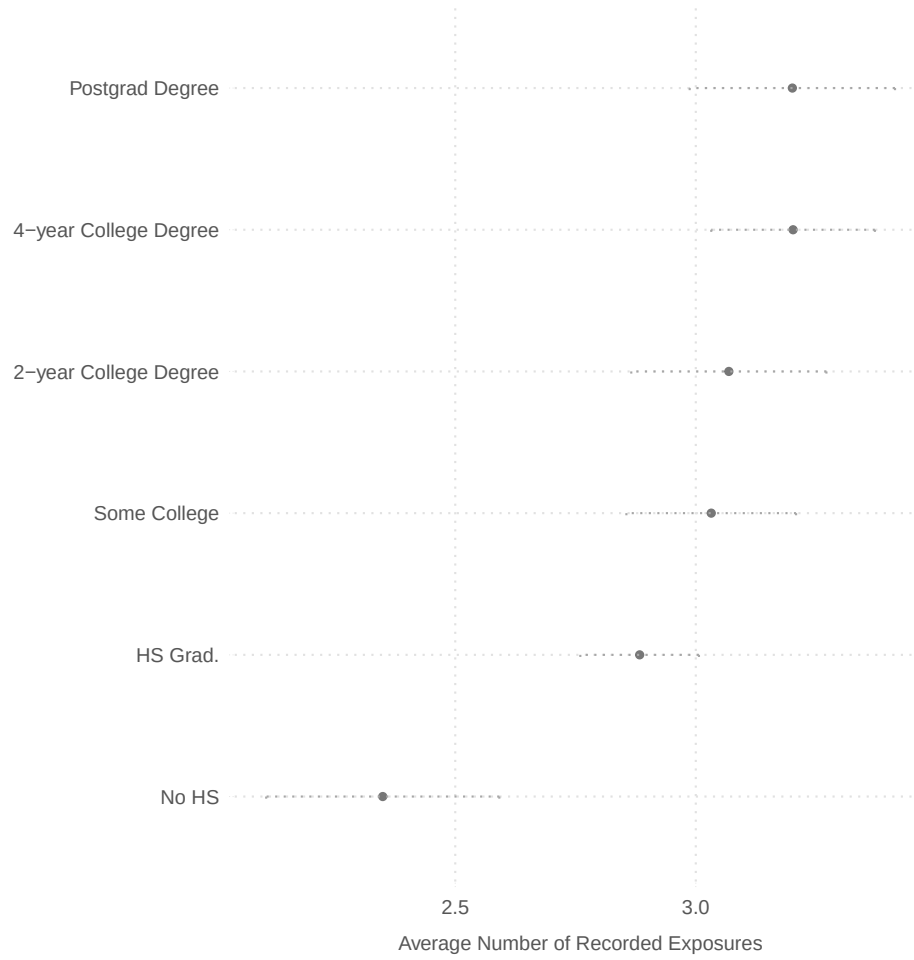


Figure SI 2.4: Relationship Between Age and Number of Breaches

